Edge-Ready ECG Classification using CRISP-DM



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Summary

This report covers the whole project, from the problem (section 1) and the data (sections 2 and 3) to modelling, evaluation and deployment (sections 4 to 6). EdgeECGNet (25,268 parameters, 25 KB in INT8) was evaluated on the 22 DS2 records, which were never used for training or tuning. The model fits the MAX78002 (synthesis passed), loses little macro-F1 from FP32 to INT8 and degrades gradually under noise, but it does not reach the macro-F1 and minority-class targets. The errors of classes S and F are concentrated in a few records and come from missing timing information (no RR interval), not from the network size.

Two INT8 models are reported, and they must not be mixed up: the PyTorch INT8 emulation is the main, pre-planned model (all noise and error analysis refer to it), and the ai8x 8-bit model is the one that was actually synthesized for the MAX78002. They come from separate training runs. Neither was chosen using DS2.

Honest reading of the numbers: a tiny 2,900-parameter CNN reaches a similar or higher macro-F1 than the edge model, so the larger model is justified by the INT8 and deployment analysis, not by accuracy.

1. Business/Application Understanding

1.1 Application scenario

What problem are you trying to solve?

The phrasing of the problem can be made by asking the question: “How do we use machine learning to detect abnormal heartbeats and classify the signals, all while still being able to be compatible with an Edge-AI device with a maximum size of 2MB?”.

Where could automatic ECG classification be useful?

Because the primary deep learning algorithms (specifically CNNs in our case) can understand short-term patterns within individual heartbeats, and long-term irregularities spanning across multiple heartbeats, these algorithms can detect conditions like premature ventricular contraction and atrial fibrillation (Ansari et al., 2023). Automatic ECG would therefore, of course, be useful in healthcare. While automatic ECG could be used for both active testing and for monitoring in a clinical setting, it could also be implemented in sensors of mobile devices and smartwatches to detect abnormal signals even outside of clinical settings, alerting individuals to seek treatment. Because a 1D CNN requires relatively low computational power, it could work as lightweight monitoring built into smart devices.

Why might inference directly on an Edge-AI device be useful?

Offline functionality, because the AI model runs locally there is no reliance on an internet-connection. This is also an important aspect in scenarios of crisis.

Then there is the aspect of data privacy. Because the data does not leave the device, it enforces the idea of privacy in the users data. Not to mention that it also makes it easier complying with data privacy laws.

Another aspect could be battery and data usage, sending a continuous ECG stream uses a lot of energy and bandwidth. A small chip like the MAX78002 can run a small neural network with very little energy.

Who could potentially use such a system?

The potential users and key stakeholders for this system includes a broad number of individuals and medical professionals. Primarily, high-risk patients and elderly populations represent a core user group. Ramkumar et al. (2018) point out that atrial fibrillation prevalence exceeds 15% in individuals over 85 years old and is strongly linked to underlying conditions such as hypertension, diabetes, and prior stroke. These individuals could benefit from accessible, low-burden continuous screening during their daily routines (Niu et al., 2023). Furthermore, clinicians and healthcare providers serve as important users, as automated diagnostic triage tools can reduce clinical diagnostic workloads, standardize signal analysis, and assist cardiologists and primary care physicians in capturing sporadic, brief arrhythmic episodes that might otherwise go undetected during standard clinical visits (Ansari et al., 2023). The system also caters to the general public and health conscious consumers and as noted by Niu et al. (2023), everyday users of commercial smartwatches and smart wearables can engage in proactive personal health tracking, enabling the early detection of hidden or asymptomatic cardiac abnormalities.

1.2 Data-Mining Objective

Input:

The input of the machine-learning model is an ECG heartbeat segment extracted from the MIT-BIH Arrhythmia Database. The segment is centered around an annotated heartbeat and consists of 360 samples (one second at 360 Hz: 180 before the R-peak, the R-peak itself and 179 after) of lead MLII, band-pass filtered at 0.5–40 Hz and normalised per beat to the range [-1, 1].

Output:

The model predicts the class of the input heartbeat. Each heartbeat is assigned to one of four classes: N (Normal-type), S (Supraventricular ectopic), V (Ventricular ectopic), or F (Fusion).

Type of task:

The problem is formulated as a supervised multiclass classification task, since the model is trained using ECG heartbeat samples with known class labels and must assign each previously unseen heartbeat to one of four possible classes.

1.3 Project Objectives

- Training of a model with a four-class heartbeat classification – N, S, V, & F.
- Create a model that will be compatible with MAX78002.
- Generalise to new ECG-recordings of people the model has never seen (test on DS2).
- Handle class imbalance, so rare classes are not ignored.
- Stay robust to ECG noise.
- Keep every step reproducible and documented.

1.4 Success Criteria

Accuracy alone is not a good measure here – if a model always answers “N”, it is right for 89% of the dataset test beats, but it will not find abnormal beats at all. So we use per class measures. The criteria below are measured on the final test set DS2. The numbers were formalised during Task 2 (the first submission described the criteria only in words); they are based on reference points (the “always N” classifier, the classical baseline and the 2 MB limit of the MAX78002) and were not used to make any modelling decision.

Area	Measure on DS2	Target
Predictive performance	Macro-F1 (always reported together with accuracy and the confusion matrix)	Macro-F1 at least 0.6 and at least 0.05 above the best classical baseline (“always N” gives about 0.24)
Minority-class performance	Recall (sensitivity) of V, S and F, with precision and F1	Recall V at least 0.8, S at least 0.5, F at least 0.5
Noise robustness	Macro-F1 under real NSTDB noise (bw, em, ma) at 12, 6 and 0 dB, compared with clean	Drop from clean to 0 dB of at most 0.05 (mean of the three noise types)
Model complexity	Size of the 8-bit model; macro-F1 change from FP32 to INT8	At most 2 MB (goal: 100,000 parameters or fewer); INT8 loses at most 0.03 macro-F1
MAX78002 compatibility	Only layers supported by the MAX78002; official ai8x synthesis	Synthesis with ai8xize passes (software-level evidence; no board is used)
Process quality	Data split and use of the test set	No record in two splits; DS2 used only for the final evaluation and every use logged

1.5 Constraints

- Limited Edge-AI memory

The limit of the Edge-AI memory is 2MB, which means that we will have to fit inside that 2MB constraint. By making the model fit into such constraints, we lose out on the opportunity for an increase in model complexity.

- Fixed-point/INT8 computation

For actual deployment into the MAX78002, the model would have to be quantized down to 8-bit for the deployment.

- Class imbalance

The dataset is skewed, with the N-class making up the majority of the heartbeats, and the F-class making up less than 1%. This prevents the use of standard accuracy metrics.

- Limited number of subjects

The number of records in the database is 48, and the number of subjects is 47. Four records contain paced beats (102, 104, 107, 217) and are excluded, so 44 records from 43 subjects are

used. Because there is such a limited number of subjects, the model heavily depends on which subjects end up in the test set, which is why we are using the (recommended) standard protocol. Not to mention that two separate records (201 and 202) are on the same subject. This is something to keep in mind to prevent testing and training on the same subject.

- Noise

Possible noise constraints can be addressed through band-pass filtering during pre-processing.

- Computational resources

Because we will be using a 1D CNN, the device will not require substantial computational power.

- Absence of physical MAX78002 hardware

Without a board, latency, energy use and real RAM use cannot be measured. Deployment feasibility is shown only at software level, with the official MAX78002 toolchain (quantization and synthesis).

1.6 Risks and Challenges

The development of an ECG heartbeat classification system presents several challenges related to data quality, model generalization, class imbalance, and hardware limitations. Identifying these risks early is important to ensure that the resulting model is reliable and evaluated fairly.

Risk	Why is it important	Proposed mitigation
Class imbalance	The four heartbeat classes (N, S, V and F) are not equally represented in the MIT-BIH Arrhythmia Database. Normal heartbeats are much more common than abnormal ones. The model could therefore achieve high overall accuracy while performing poorly on rare but clinically relevant classes.	The proposed mitigation is to try weighted sampling or class weights to help the model learn from the less common heartbeat classes.
Data leakage and poor generalization	Heartbeats from the same patient can share similar characteristics. If beats from one patient appear in both training and test sets, the model may learn patient-specific patterns rather than generalizable heartbeat characteristics. This can produce overly optimistic performance estimates.	The proposed mitigation is to split the ECG records, so that no patient appears in both training and test sets.
Noise and signal variability	ECG recordings may contain baseline wander, muscle noise, electrode movement artefacts, and differences in signal amplitude or morphology. These variations can make it more difficult to distinguish between heartbeat classes and may reduce the model's robustness when processing real-world signals.	The proposed mitigation is to try filtering, normalization and noise augmentation to see if they help the model handle noisy ECG signals.
Hardware and computational limitations	A model that performs well in Python may be too large or use unsupported operations for deployment on the MAX78002.	The proposed mitigation is to keep the model relatively small and test whether it works with the MAX78002 toolchain.

These risks are considered throughout the project. In particular, preventing data leakage and evaluating minority-class performance are essential for obtaining meaningful results, while hardware compatibility must be considered alongside predictive performance to meet the project's Edge-AI objective.

2. Data Understanding

2.1 Dataset Description

The name of the database is MIT-BIH Arrhythmia Database. The database contains 48 ~30 minute two-channel recordings of 47 subjects (Moody & Mark, 2001). The recordings have a sampling frequency of 360 Hz per channel. Two or more cardiologists annotated each record, with a total of approximately 110000 annotations in the database. The ECG leads that are available in the dataset are MLII and V1-5. We use 44 of the 48 records (43 subjects, 103,724 expert annotations); the four records with paced beats are left out, as is common practice for this task.

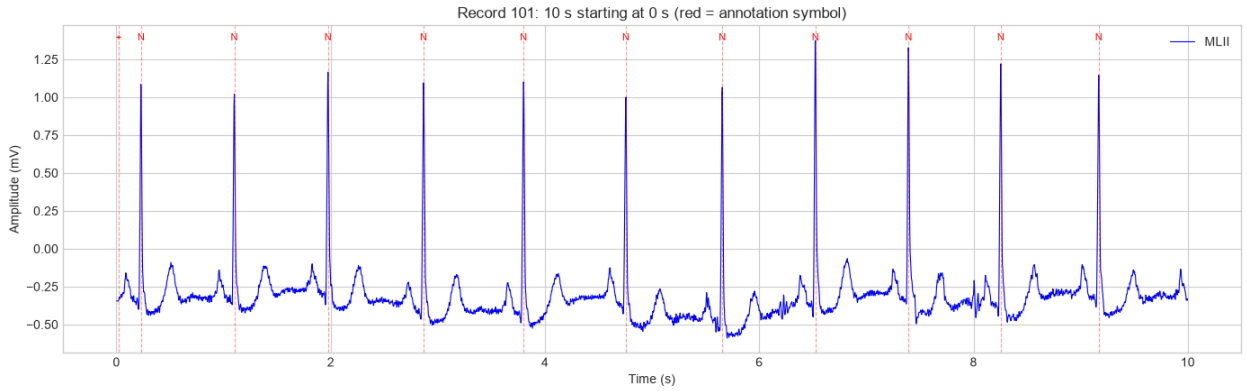
2.2 Target Classes

Final class	Original annotations included	Number of beats (all annotations)	Number of beats (final dataset)
N	N, L, R, e, j	90,125	90,076
S	S, A, a, J	2,781	2,781
V	V, E	7,009	7,008
F	F	803	802
Total		**100,718**	**100,667**

Annotations excluded are the paced beats “/” and “f”, which belong to the paced records that are not used, “Q” (unclassifiable beat, 15 beats) and the non-beat marks + ~ ! " x | [ ] (2,991 marks). The last column is 51 beats lower than the first, because their one-second window would reach outside the recording. All exclusions are counted in a file written by the preparation code.

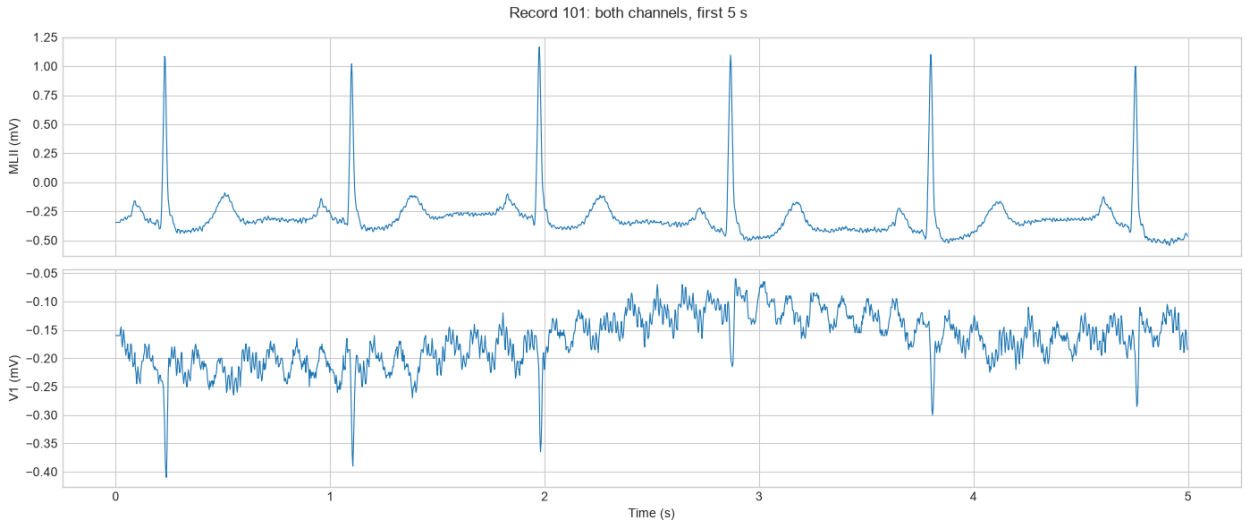
2.3 Exploratory Data Analysis

Example ECG signal:



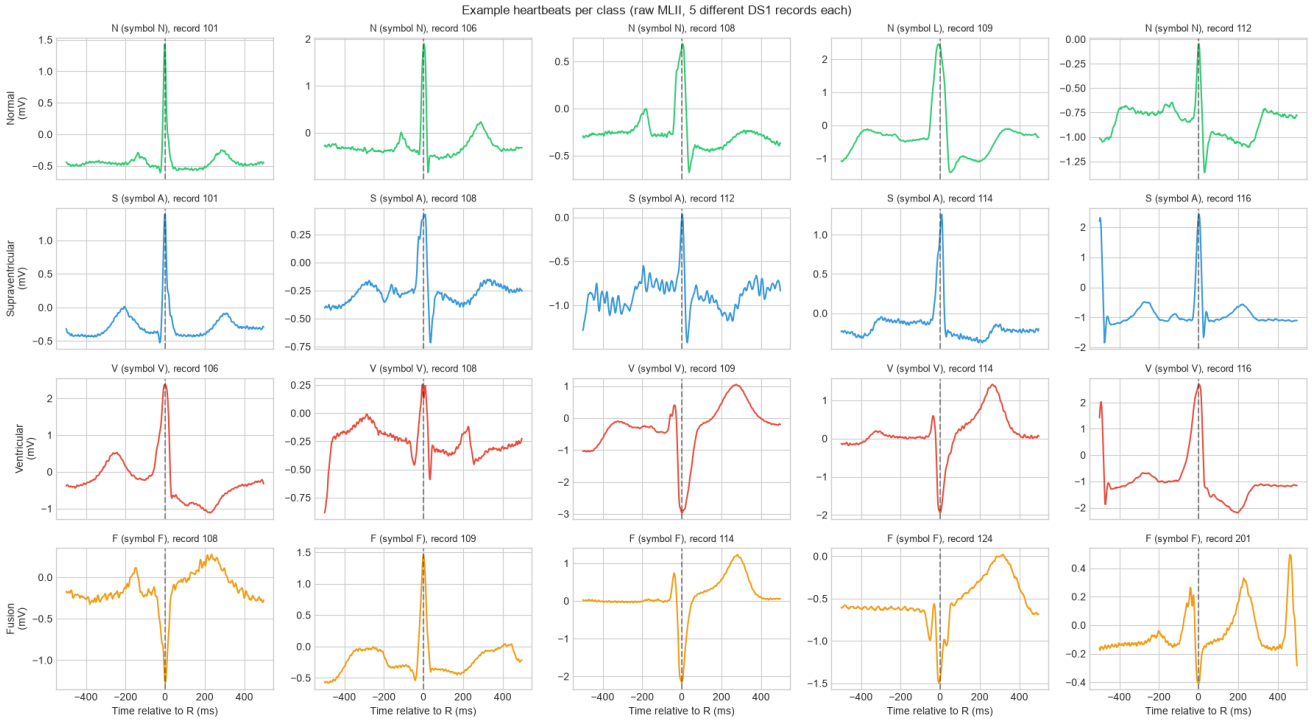
Example ECG signal

The two channels of one record:



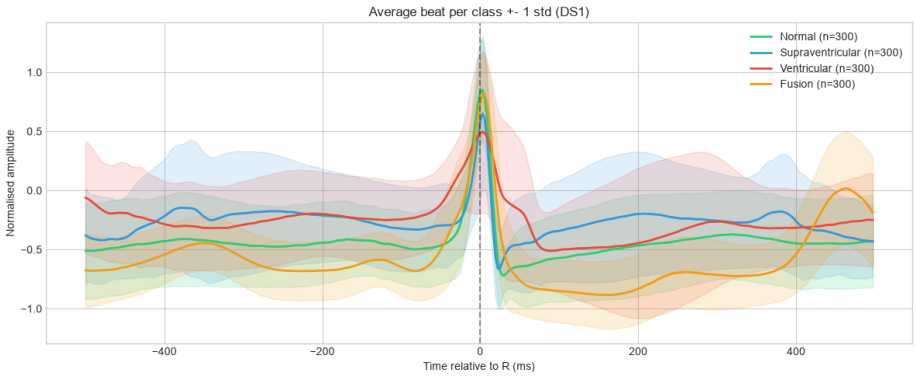
Both channels of record 101

Example heartbeat from each class:



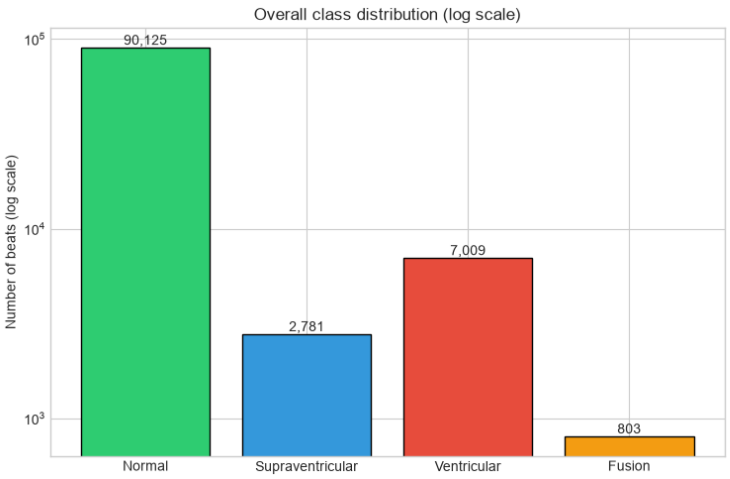
Example heartbeats

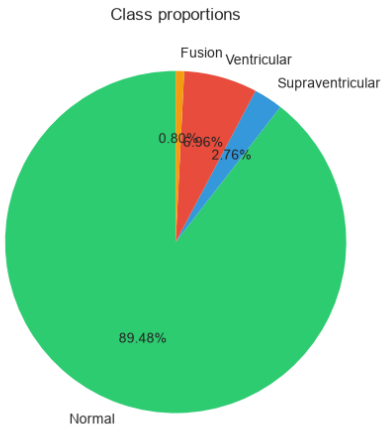
Average beat of each class:



Average beat per class

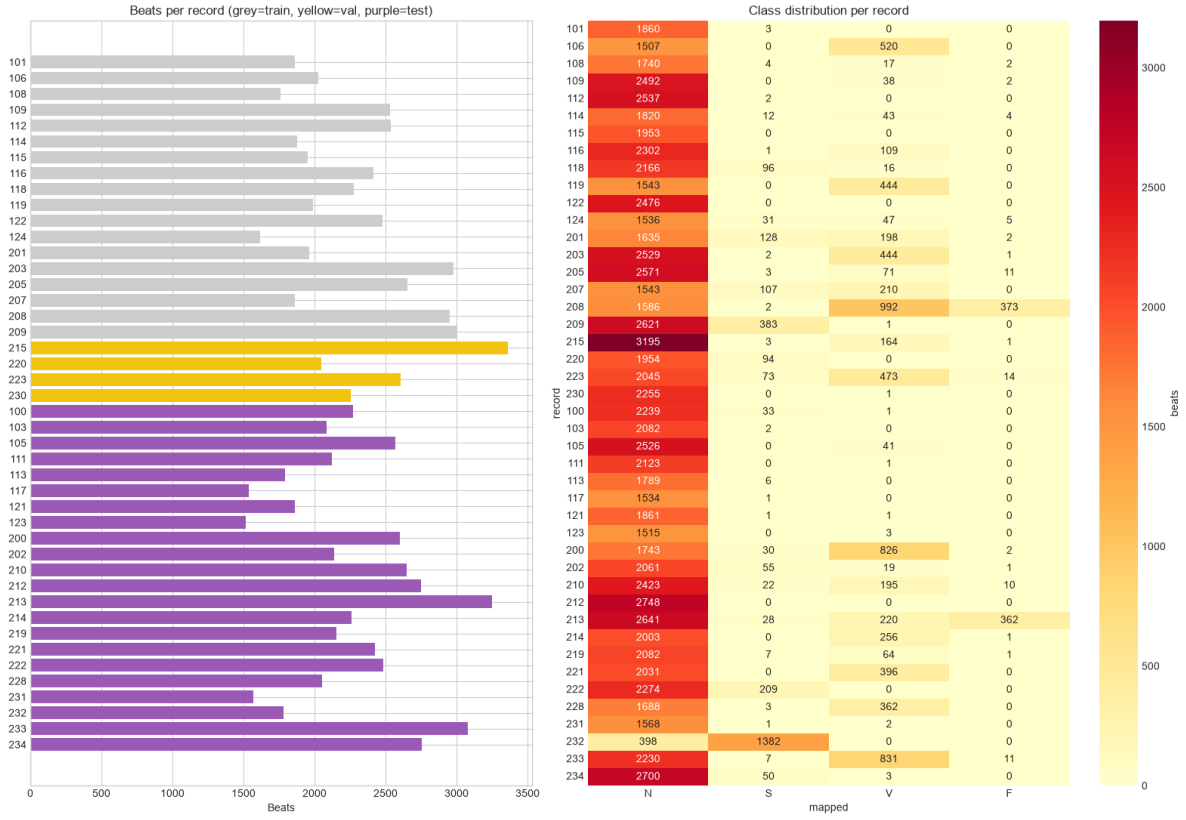
Class distribution:





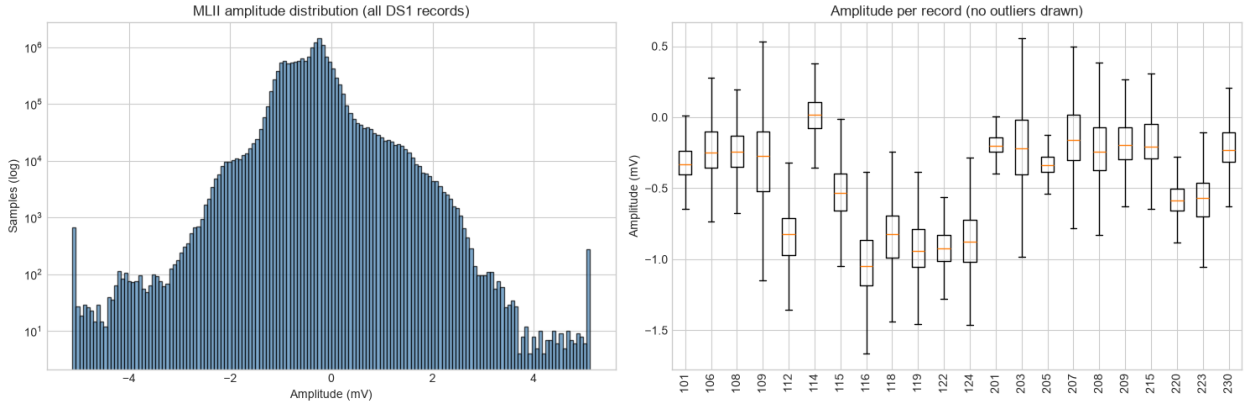
Class distribution

Subject/record distribution:



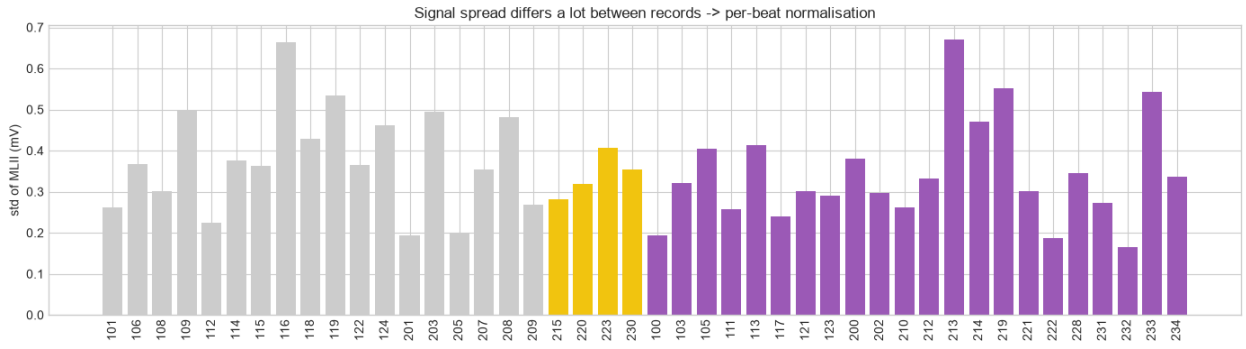
Records

Signal amplitude distribution:



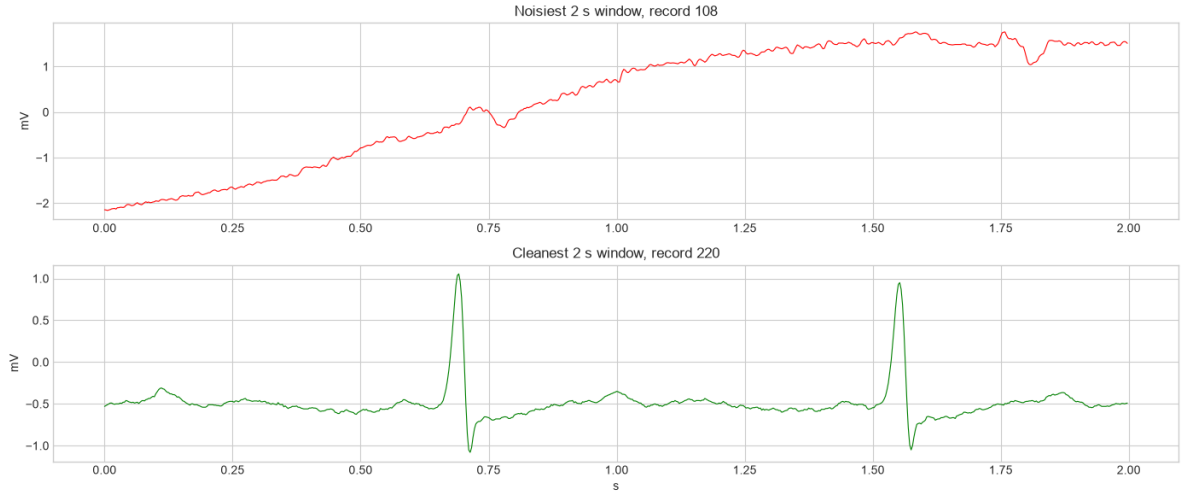
Amplitude

Differences between recordings:



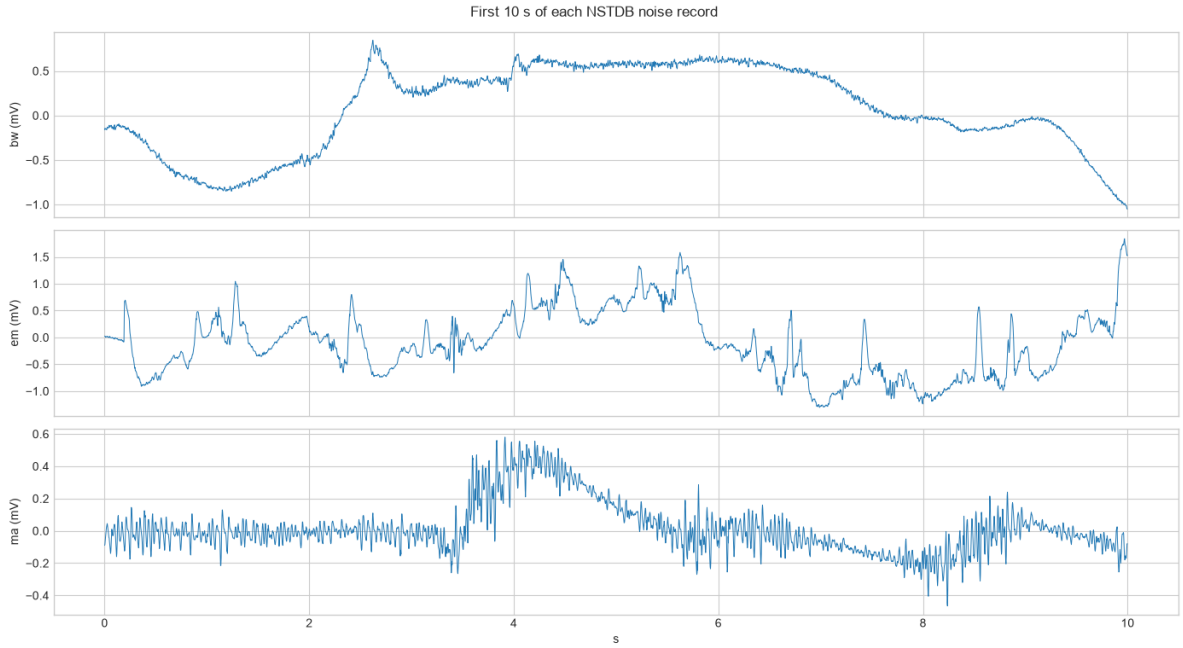
Signal spread per record

Example of noisy ECG:

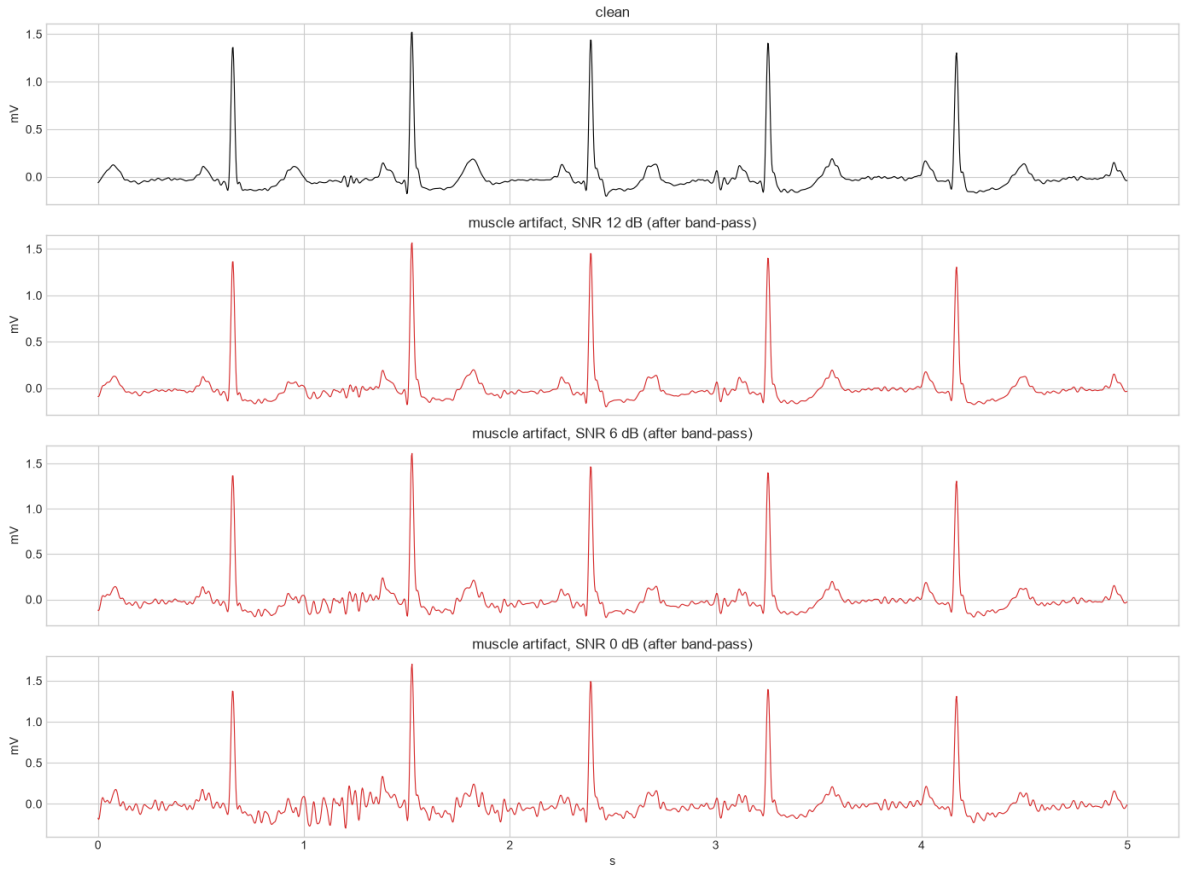


Noisiest and cleanest window

The MIT-BIH Noise Stress Test Database at 12, 6 and 0 dB:



Noise types



Muscle artifact at three noise levels

2.4 Data Quality

The data quality of the MIT-BIH Arrhythmia Database was investigated by examining missing values, extreme amplitudes, corrupted segments, class imbalance, differences between recordings and ECG channels, and noisy regions.

Missing data

The exploratory data analysis found no NaN values in the selected ECG recordings. However, this check does not rule out all forms of signal loss or recording errors.

Extreme amplitudes

Since the amplitude varies heavily between records, per-beat min-max normalisation is needed to evaluate the shape of the wave instead of the absolute voltage. The peaks at about ±5 mV in the amplitude histogram come mainly from record 116, which touches the limit of the recorder (977 samples beyond 5 mV).

Corrupted segments

Certain segments contain noise or signal loss, which is addressed through flat-beat rejection (no flat beats were found in the selected records).

Class imbalance

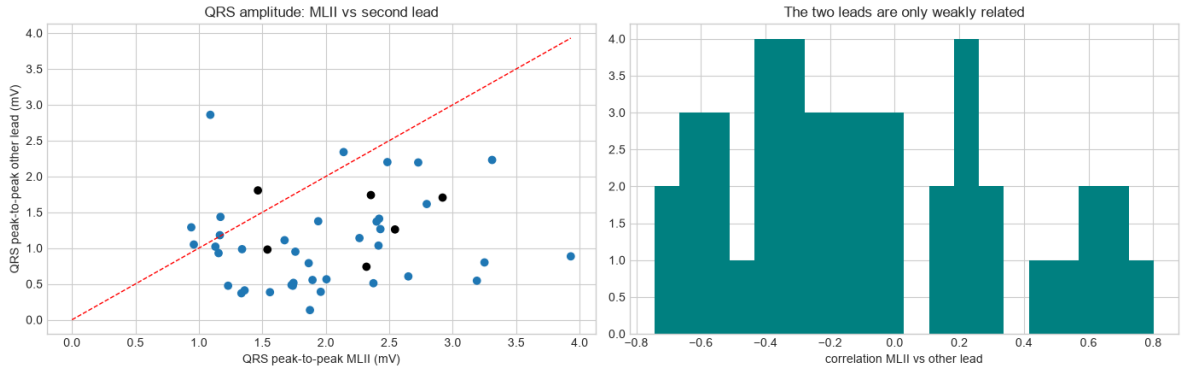
The dataset contains many more normal heartbeats than abnormal ones. Normal heartbeats make up approximately 89.5% of the data, while fusion heartbeats make up only 0.8%. This imbalance could make it harder for the model to learn to recognise the less common classes. Therefore, it is important to evaluate how well the model performs on each class, rather than only looking at its overall accuracy.

Differences between recordings

In one record (114), MLII is the second channel. Amplitudes also heavily vary between records.

Differences between ECG channels

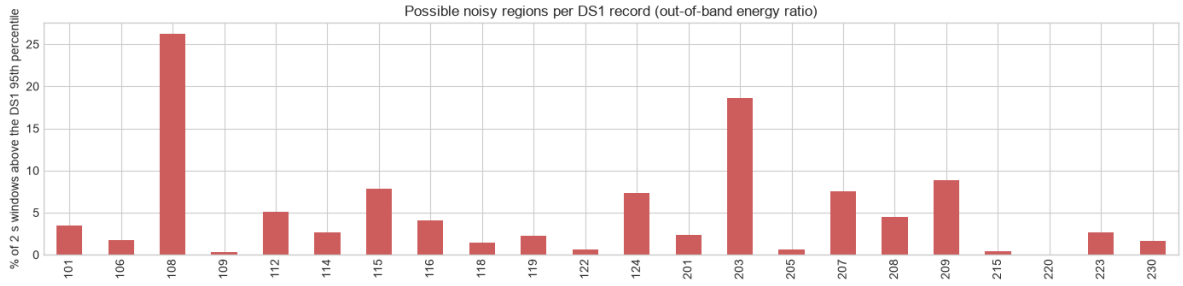
One of the ECG channels is more sensitive than the other. R-peaks also look different depending on the channel.



Differences between the two channels

Possible noisy regions

There are some noisy regions, which will be taken care of through the Butterworth band-pass filter (0.5-40Hz). Experts marked noise or artefacts in 38 of the 44 records; records 108 and 203 show the strongest baseline drift.

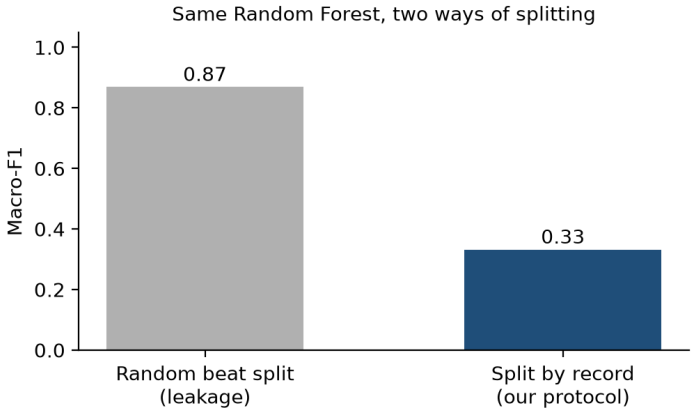


Noisy regions per record

2.5 Leakage Analysis

Data leakage could occur in several ways, one of them being if we were to randomly split beats. Beats of the same patient share heart geometry, electrode position, gain, baseline level and often the same abnormal morphology repeated hundreds of times (e.g. one patient has hundreds of near-identical V beats). If beats of one patient are in both train and test, the network can recognise the patient instead of the beat type. This results in test accuracy becoming unrealistically high, and does not predict performance on new patients. Our approach to this is using DS1 (22 records) for training and validation, split by record (18 train / 4 validation). DS2 (22 records) is the final test set, and no record appears in both sets.

We checked the size of this effect with a small experiment in the notebook: the same Random Forest reached a macro-F1 of 0.87 when the beats were split randomly and only 0.33 when the records were split.



Leakage experiment

Leakage could also occur from neighbouring beats overlapping, i.e., a one second window around R overlaps with the windows of the previous and next beat. If we were to randomly split, this can result in almost identical samples in both training and testing.

The database has 48 records but 47 subjects, records 201 (DS1) and 202 (DS2), which are records from the same individual. The standard DS1/DS2 split therefore has one subject in both sets. We keep the standard split because the task requires it, and report it as a known limitation.

Normalisation computed on the whole dataset would also leak test statistics. Here, normalisation is per beat, and class weights are only computed on the training set. Validation must also be split by record, and augmentation is only applied to training data. Noise for training and test comes from different halves of the NSTDB noise records.

3. Data Preparation

3.1 Record Selection

For the record selection, the commonly used inter-patient MIT-BIH partitioning is used. This means that the records used for training and validation are: 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, & 230.

For the final test, the records used are: 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, & 234. The four records with paced beats (102, 104, 107, 217) are not used in any split.

3.2 ECG Channel Selection

We are using the MLII channel primarily because it is the standard lead for the MIT-BIH classification, but also for clean preprocessing and an easier detection of the heartbeat. The preprocessing is much easier because the R-peaks are much more prominent in the MLII channel.

Because of a single record, where MLII is the second channel, MLII is instead selected by name. If a record does not have MLII at all, it will fall back to channel 0 and report it.

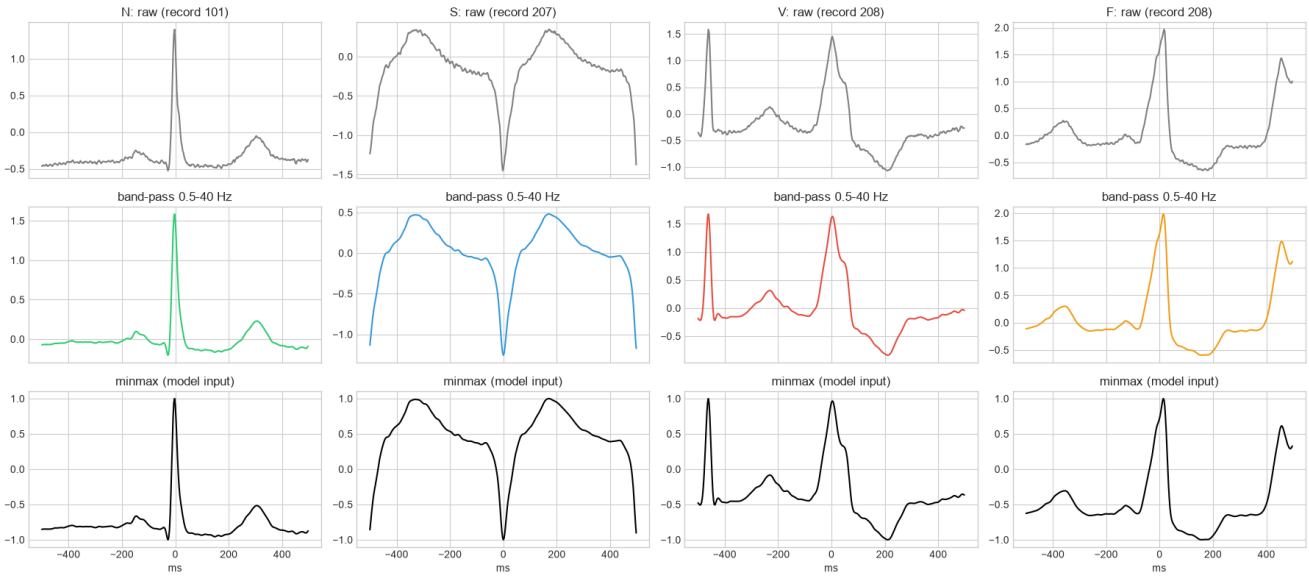
3.3 Beat Extraction

- How R-peak annotations are obtained: they are read directly from the .atr-files.
- Samples before the annotated beat: 180.
- Samples after the annotated beat: 179.
- Final input length: 360 samples.

3.4 Signal Preprocessing

Raw ECG Data with MLII lead
↓
Noise injection (only for the noise test, added to the raw signal)
↓
Band-pass filtering (0.5-40Hz, Butterworth, order 4, zero-phase)
↓
Beat segmentation around annotated R-peaks
↓
Drop flat-line beats
↓
Normalization per beat (min-max to [-1, 1])
↓
Model Input

The zero-phase filter uses samples from the future, so it cannot run in real time. A device would need a causal filter; the preparation code has an option for this, but it was not used for the results.



Preprocessing of one beat per class: raw, band-pass filtered, normalised

3.5 Label Preparation

N (Normal): N → N, L → N, R → N, e → N, j → N

S (Supraventricular): S → S, a → S, A → S, J → S

V (Ventricular): V → V, E → V

F (Fusion): F → F

3.6 Class Imbalance

The MIT-BIH dataset contains a severe class imbalance. The N class contains the majority of the heartbeats, 90,125, for the total 100,718 heartbeats (90,076 of 100,667 in the final dataset). That is why, with the usage of class weights, forcing the model to prioritize minority classes is a necessity. The weights are computed from the training set only (N 0.28, S 13.15, V 3.23, F 25.51). In the model selection of Task 2 a softer version, the square root of these weights, gave the best validation macro-F1 (0.514 against 0.513 for the full weights) and is used for the final model.

The following alternatives were rejected:

- Oversampling: random oversampling would cause overfitting.
- Undersampling: undersampling would starve the model. The model needs a big volume of heartbeats to learn what a healthy baseline looks like.
- Balanced batches: balanced batches would require the same small amount of F beats. This would result in the model learning about those specific patients instead of learning generalizable features. In the Task 2 model selection a balanced sampler was also tested and was worse on the validation set (macro-F1 0.486 against 0.514 for softened class weights).
- Focal loss: the class weights provide a more simple and interpretable baseline than focal loss, and without needing additional hyperparameter tuning.
- No balancing: without balancing, the model would predict the N-class for every input, achieving high accuracy while not achieving its objective of detecting dangerous arrhythmias.

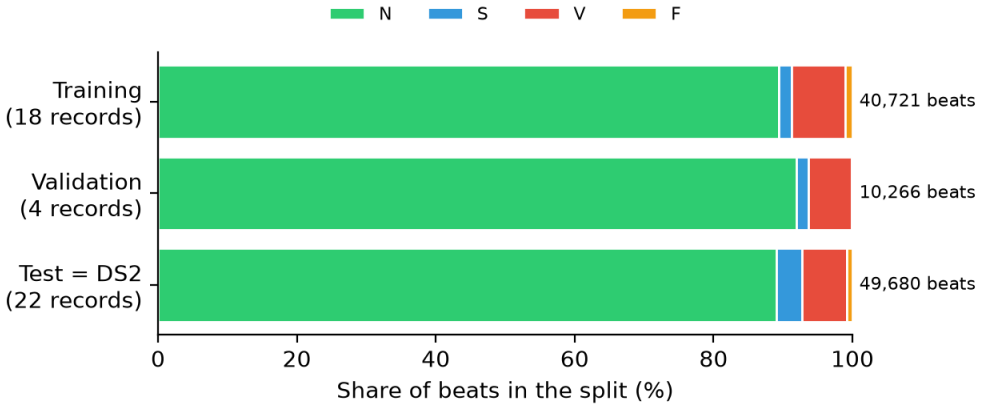
3.7 Data Augmentation

To prevent the model from overfitting from the limited pool of subjects, and to simulate real-world conditions for the Edge-AI deployment, noise injection and baseline wander injection was used as augmentation on our training data, together with a random gain (0.9–1.1) and a small time shift (up to 3 samples). Augmentation is applied to training beats only. The noise comes from the first half of the NSTDB recordings; the second half is kept for the DS2 noise test.

3.8 Final Dataset

Split	Records	N	S	V	F	Total
Training	18	36,398	774	3,150	399	40,721
Validation	4	9,443	170	638	15	10,266
Test	22	44,235	1,837	3,220	388	49,680
All	44	90,076	2,781	7,008	802	100,667

The three splits have a similar class mix; the validation set contains only 15 F beats, so validation results for F are noisy.



Class mix of the three splits

3.9 Reproducibility

Software versions: Python (3.9.13), NumPy (1.26.4), SciPy (1.9.1), pandas (1.4.4), & WFDB (4.1.2). Task 2 (modelling, evaluation, report) used Python 3.12.3, PyTorch 2.14.1 (run on CPU), NumPy 2.5.4, SciPy 1.18.1, pandas 3.1.0 and WFDB 4.3.1. For the MAX78002 toolchain a separate environment was used, with Python 3.11, PyTorch 2.3.1 (CPU) and NumPy 1.26.4, together with Analog Devices' ai8x-training and ai8x-synthesis.

Random seed: 42.

Preprocessing parameters:

- Target lead: MLII
- Filter: Butterworth band-pass (lowcut Hz: 0.5, highcut Hz: 40.0, order: 4, zero phase: true).
- 180 samples before R, 179 samples after R.
- Normalization: Min-max.
- Pipeline order: Noise injection (noise test only), band-pass filter, optional clipping, segmentation, flat-beat rejection, per-beat normalisation.

Sampling frequency: 360Hz.

Input size: 360 samples.

3.10 Summary, Limitations and Next Steps

What we learned about the data:

- The data is clean (no missing values, no flat parts), but it has a strong class imbalance and large differences between records.
- The lead order is not the same in all records; selecting MLII by name solves this.
- Experts marked noise or artifacts in 38 of 44 records, and some records have strong baseline drift.
- A random beat split gives far too optimistic results.

Limitations

- Records 201 (train) and 202 (test) are the same person.
- Only 15 F beats in the validation set; validation metrics for F and S are noisy.
- Beats near expert-marked noisy regions are kept in the data.
- R-peak positions come from expert annotations, not from an automatic detector.
- The zero-phase filter cannot be used in real time, and one second of ECG contains no information about the time between beats (RR interval).
- No claim of clinical validity.

Next steps (CRISP-DM stage 4: Modeling)

- Build a simple baseline and a compact 1D CNN that fits the MAX78002 rules.
- Train with class weights (and compare with the balanced sampler), using the training set; select with the validation set.
- Report per-class results; run the noise test at 12, 6 and 0 dB; compare FP32 and INT8.

4. Modeling

4.2 Edge model: EdgeECGNet

Layer	Input	Operation	Output	Parameters
conv1	1 x 360	Conv1d(k=5, pad=2) + BN(folded) + ReLU	16 x 360	96
conv2	16 x 360	MaxPool1d(2) -> Conv1d(k=5, pad=2) + BN(folded) + ReLU	32 x 180	2,592
conv3	32 x 180	MaxPool1d(2) -> Conv1d(k=5, pad=2) + BN(folded) + ReLU	32 x 90	5,152
conv4	32 x 90	MaxPool1d(2) -> Conv1d(k=3, pad=1) + BN(folded) + ReLU	48 x 45	4,656
conv5	48 x 45	MaxPool1d(2) -> Conv1d(k=3, pad=1) + BN(folded) + ReLU	64 x 22	9,280
conv6	64 x 22	MaxPool1d(2) -> Conv1d(k=1, pad=0) + BN(folded) + ReLU	32 x 11	2,080
fc	32 x 11 = 352	Flatten -> Linear (32-bit output)	4	1,412

Deployed parameters: 25,268 (INT8 weights 25,268 bytes); estimated peak activation memory 11,520 bytes.

4.3 Training configuration

Setting	Value
Architecture	edge_cnn (width 1.0, dropout 0.1)
Optimizer / learning rate	adam / 0.001 (cosine schedule), weight decay 0.0
Batch size / epochs	128 / 30
Loss	cross_entropy, label smoothing 0.0
Class weighting	class_weights, power 0.5
Augmentation	True
Early stopping	patience 30, min epochs 10; selection from epoch 1
Gradient clipping	1.0
QAT	30 epochs, lr factor 0.1, patience 12
Seed	42

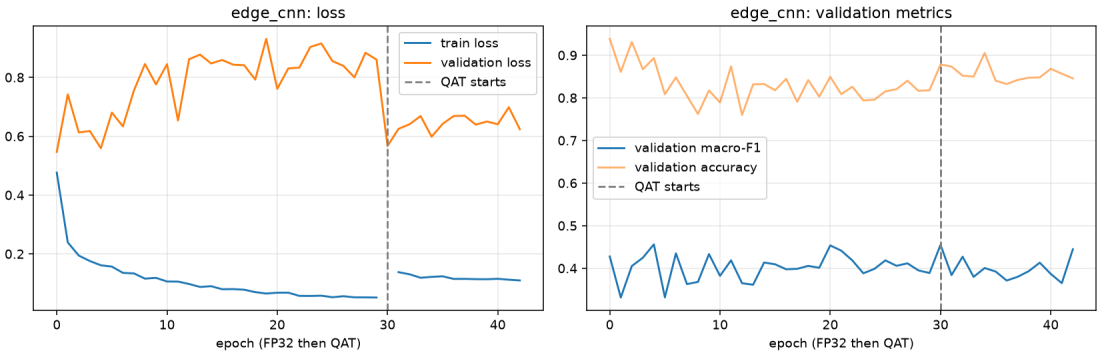
FP32 best validation macro-F1 0.4560 at epoch 5; QAT best 0.4551 at epoch 0 (epoch 0 means QAT did not improve on the converted FP32 model, so QAT equals PTQ); training time 519.9 s on cpu.

4.4 Model selection (validation macro-F1 only)

Experiment	Overrides	Val macro-F1	Best epoch	Parameters	Selected
tiny_cnn	{'model': {'arch': 'tiny_cnn', 'dropout': 0.0}}	0.4584	2	2900	
edge_w0.5	{'model': {'width_mult': 0.5}}	0.4533	2	6748	
edge_w1.0	{'model': {'width_mult': 1.0}}	0.5127	11	25268	yes
edge_w1.5	{'model': {'width_mult': 1.5}}	0.4658	5	55564	
edge_w1.0_lr3e-3	{'training': {'lr': 0.003}}	0.4834	4	25268	
edge_w1.0_sampler	{'training': {'imbalance': 'balanced_sampler'}}	0.4855	9	25268	
edge_w1.0_softweights	{'training': {'class_weight_power': 0.5}}	0.5140	11	25268	

Iterative runs after the sweep (validation only)

Run	Change	Result	Decision
D	dropout 0.1, weight decay 0, class-weight power 0.5	val macro-F1 0.4560 FP32 / 0.4551 INT8	frozen as best_config.yaml
E	dropout 0.3, weight decay 1e-4	no validation gain (see logs)	rejected



Training curves of the final edge model (validation records).

5. Evaluation

5.1 Overall results

Metric	Baseline (best classical)	Tiny CNN (FP32)	Edge model FP32	INT8 post-training	INT8 QAT (emulated)
Accuracy	0.8956	0.9204	0.9035	0.8967	0.8967
Macro F1	0.4012	0.4970	0.4145	0.4019	0.4019
Model parameters	208,796	2,900	25,268	25,268	25,268

Baseline parameter count: Logistic Regression = weights + biases; Random Forest = total tree nodes.

5.2 Per-class performance

Baseline (best classical)

Class	Precision	Sensitivity/Recall	F1	Support
N	0.9347	0.9511	0.9429	44235
S	0.7692	0.0054	0.0108	1837
V	0.5728	0.7478	0.6487	3220
F	0.0022	0.0026	0.0024	388

Tiny CNN (FP32)

Class	Precision	Sensitivity/Recall	F1	Support
N	0.9483	0.9737	0.9608	44235
S	0.1524	0.0844	0.1086	1837
V	0.9154	0.7630	0.8323	3220
F	0.0731	0.1057	0.0864	388

Edge model FP32

Class	Precision	Sensitivity/Recall	F1	Support
N	0.9568	0.9583	0.9576	44235
S	0.1385	0.0441	0.0669	1837
V	0.5312	0.7475	0.6211	3220
F	0.0156	0.0103	0.0124	388

INT8 post-training

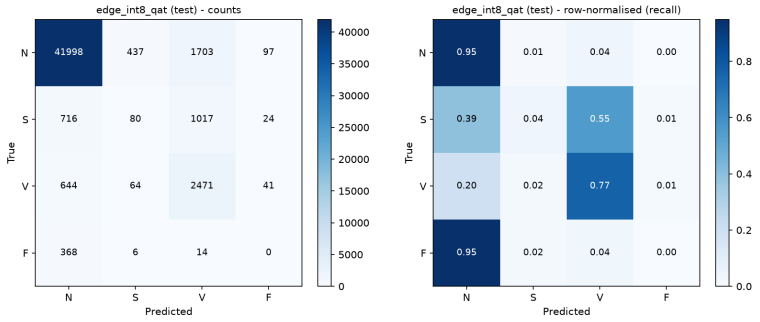
Class	Precision	Sensitivity/Recall	F1	Support
N	0.9605	0.9494	0.9549	44235
S	0.1363	0.0435	0.0660	1837
V	0.4747	0.7674	0.5866	3220
F	0.0000	0.0000	0.0000	388

INT8 QAT (emulated)

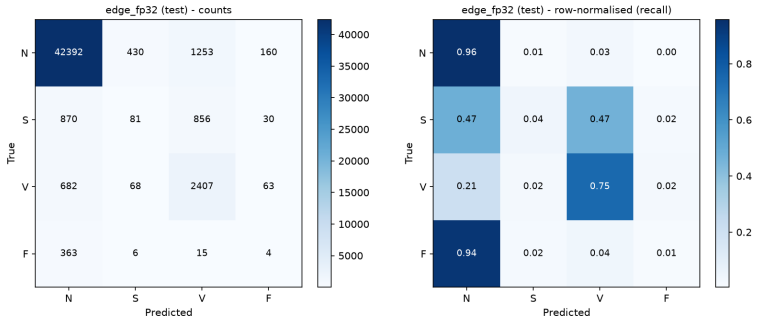
Class	Precision	Sensitivity/Recall	F1	Support
N	0.9605	0.9494	0.9549	44235
S	0.1363	0.0435	0.0660	1837
V	0.4747	0.7674	0.5866	3220
F	0.0000	0.0000	0.0000	388

5.3 Quantization analysis

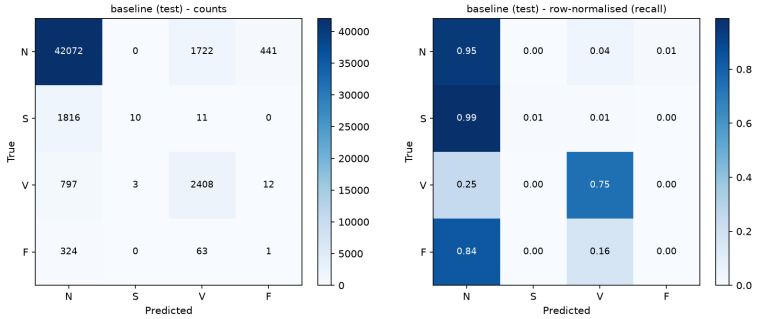
- FP32 macro F1: 0.4145 | INT8 post-training (PTQ) macro F1: 0.4019 | difference: -0.0126
- FP32 macro F1: 0.4145 | INT8 QAT (emulated) macro F1: 0.4019 | difference: -0.0126



Confusion matrix, INT8 model.



Confusion matrix, FP32 model.



Confusion matrix, classical baseline.

5.4 Noise robustness

baseline: mean over the noise types (bw, em, ma)

Condition	Accuracy	Macro F1	F1 N	F1 S	F1 V	F1 F
Clean	0.8956	0.4012	0.943	0.011	0.649	0.002
12 dB	0.8928	0.3978	0.941	0.009	0.640	0.002
6 dB	0.8903	0.3933	0.940	0.009	0.623	0.002
0 dB	0.8862	0.3852	0.937	0.005	0.596	0.003

tiny_cnn_fp32: mean over the noise types (bw, em, ma)

Condition	Accuracy	Macro F1	F1 N	F1 S	F1 V	F1 F
Clean	0.9204	0.4970	0.961	0.109	0.832	0.086
12 dB	0.9105	0.4868	0.955	0.101	0.815	0.076
6 dB	0.8951	0.4711	0.946	0.093	0.783	0.063
0 dB	0.8717	0.4439	0.932	0.083	0.713	0.048

edge_fp32: mean over the noise types (bw, em, ma)

Condition	Accuracy	Macro F1	F1 N	F1 S	F1 V	F1 F
Clean	0.9035	0.4145	0.958	0.067	0.621	0.012
12 dB	0.8938	0.4057	0.952	0.062	0.596	0.012
6 dB	0.8831	0.3988	0.946	0.060	0.578	0.011
0 dB	0.8641	0.3879	0.933	0.056	0.547	0.015

edge_int8_ptq: mean over the noise types (bw, em, ma)

Condition	Accuracy	Macro F1	F1 N	F1 S	F1 V	F1 F
Clean	0.8967	0.4019	0.955	0.066	0.587	0.000
12 dB	0.8879	0.3954	0.950	0.063	0.566	0.002
6 dB	0.8773	0.3895	0.944	0.060	0.550	0.004
0 dB	0.8584	0.3799	0.931	0.056	0.524	0.009

edge_int8_qat: mean over the noise types (bw, em, ma)

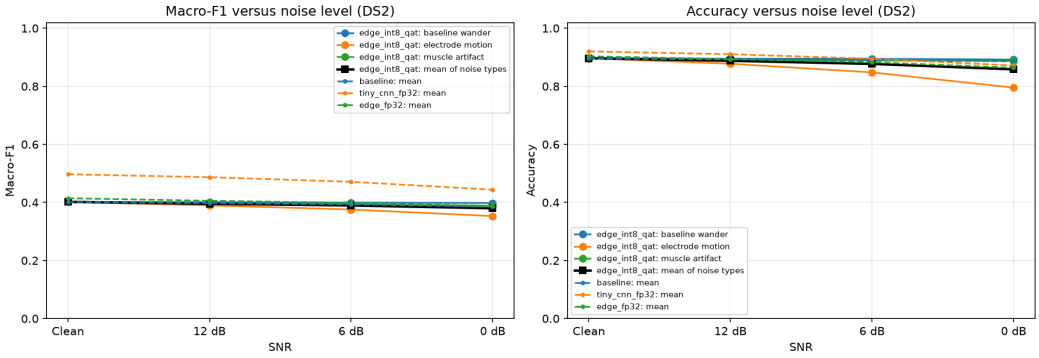
Condition	Accuracy	Macro F1	F1 N	F1 S	F1 V	F1 F
Clean	0.8967	0.4019	0.955	0.066	0.587	0.000
12 dB	0.8879	0.3954	0.950	0.063	0.566	0.002
6 dB	0.8773	0.3895	0.944	0.060	0.550	0.004
0 dB	0.8584	0.3799	0.931	0.056	0.524	0.009

edge_int8_qat: per noise type

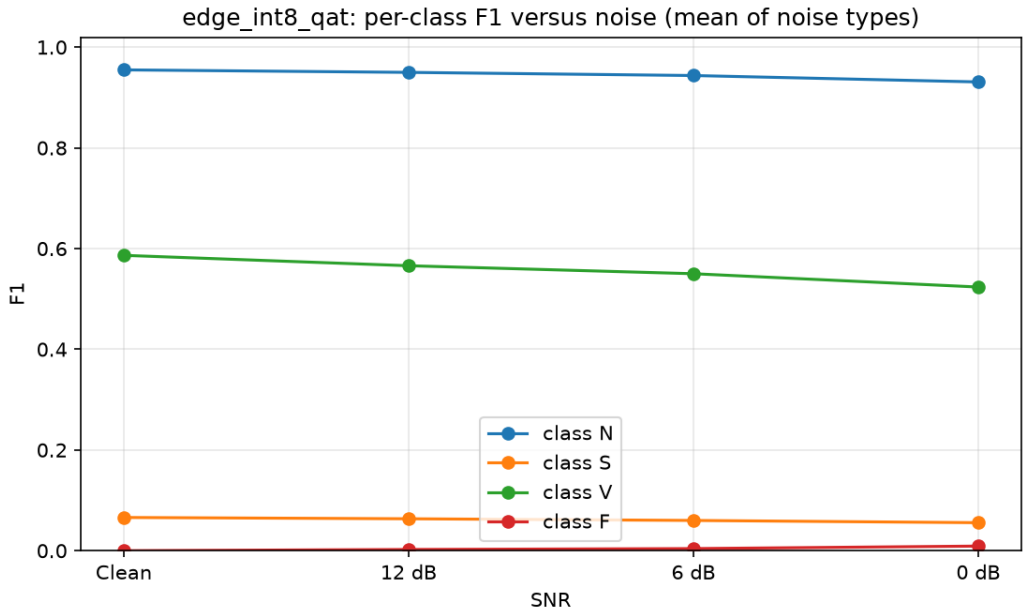
Noise type	SNR	Accuracy	Macro F1
baseline wander (bw)	12 dB	0.8953	0.4007
baseline wander (bw)	6 dB	0.8948	0.3997
baseline wander (bw)	0 dB	0.8923	0.3980
electrode motion (em)	12 dB	0.8779	0.3898
electrode motion (em)	6 dB	0.8480	0.3756
electrode motion (em)	0 dB	0.7955	0.3530
muscle artifact (ma)	12 dB	0.8906	0.3959
muscle artifact (ma)	6 dB	0.8890	0.3932
muscle artifact (ma)	0 dB	0.8874	0.3887

Band-pass filter ablation (main model, mean macro-F1)

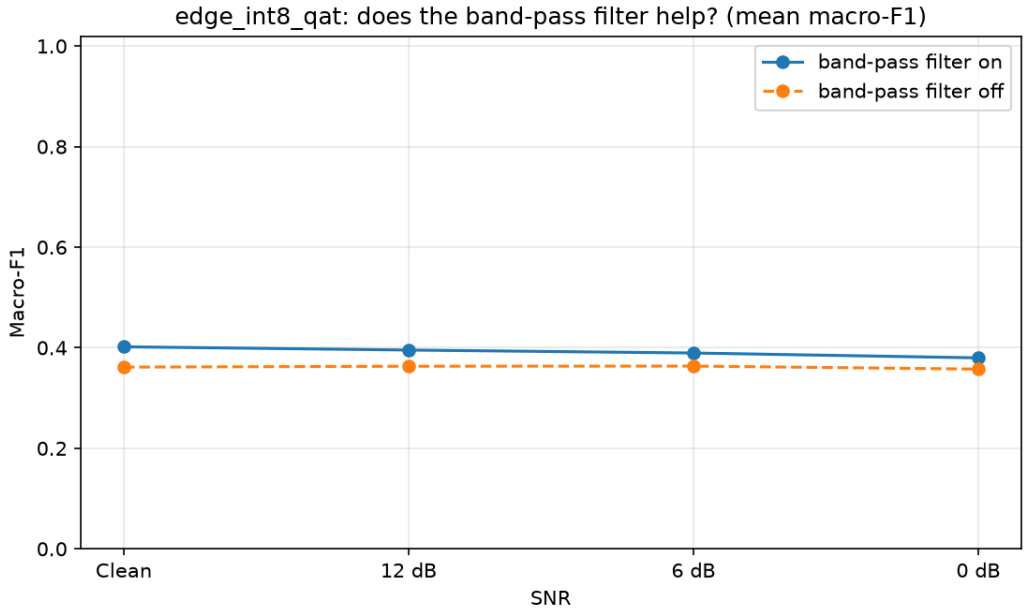
Condition	filter ON	filter OFF
Clean	0.4019	0.3614
12 dB	0.3954	0.3632
6 dB	0.3895	0.3635
0 dB	0.3799	0.3575



Accuracy and macro-F1 versus SNR.



Per-class F1 versus SNR.



Band-pass filter on versus off.

5.5 Error analysis

Beats: 49680 | accuracy 0.8967 | macro-F1 0.4019 | errors: 5131

1. Most frequent confusions

True	Predicted	Beats	% of the true class
N	V	1703	3.8%
S	V	1017	55.4%
S	N	716	39.0%
V	N	644	20.0%
N	S	437	1.0%
F	N	368	94.8%
N	F	97	0.2%
V	S	64	2.0%

- Class S: 4.4% recall; 58% of its errors are predicted as V.
- Class V: 76.7% recall; 86% of its errors are predicted as N.
- Class F: 0.0% recall; 95% of its errors are predicted as N.

2. Error rate per original MIT-BIH symbol

Symbol	Class	Beats	Errors	Error rate	Most common wrong class
N	N	36423	720	2.0%	S
L	N	4124	1057	25.6%	V
R	N	3475	448	12.9%	V
V	V	3219	748	23.2%	N
A	S	1736	1673	96.4%	V
F	F	388	388	100.0%	N
j	N	213	12	5.6%	V
J	S	51	51	100.0%	N
a	S	50	33	66.0%	N
E	V	1	1	100.0%	N

3. Records with the highest error rate

Record	Beats	Errors	Error rate	Class counts N/S/V/F
232	1780	1601	89.9%	398/1382/0/0
213	3249	644	19.8%	2639/28/220/362
111	2124	576	27.1%	2123/0/1/0
214	2258	574	25.4%	2001/0/256/1
222	2481	443	17.9%	2272/209/0/0
228	2052	331	16.1%	1687/3/362/0
212	2747	227	8.3%	2747/0/0/0
233	3077	184	6.0%	2229/7/830/11

The 3 records with most errors hold 55% of all errors.

4. Example figures

- `error_examples_N_to_S.png`: True N predicted as S (false alarms for S) (437 beats in total)
- `error_examples_S_to_N.png`: True S predicted as N (missed S) (716 beats in total)
- `error_examples_V_missed.png`: True V that the model missed (749 beats in total)
- `error_examples_F_missed.png`: True F that the model missed (388 beats in total)
- `error_mean_beats.png`: average beat of each class, correctly classified versus misclassified

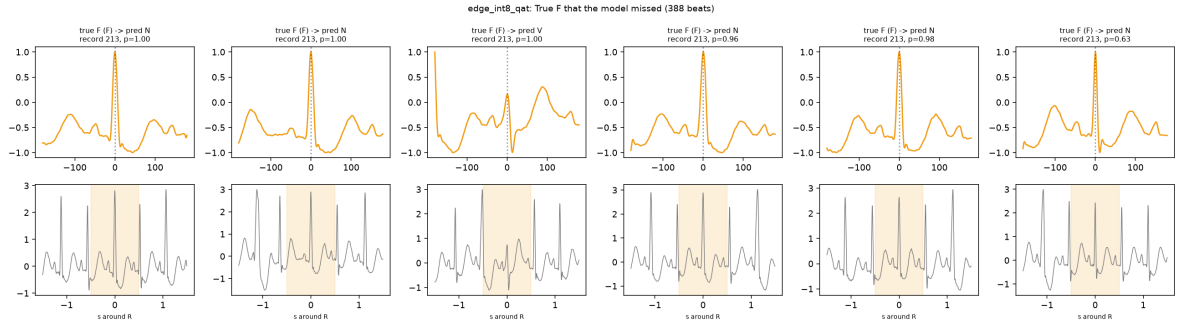
5. Effect of noise (ma, 0 dB SNR)

Clean: accuracy 0.8967, macro-F1 0.4019. Noisy: accuracy 0.8874, macro-F1 0.3887.

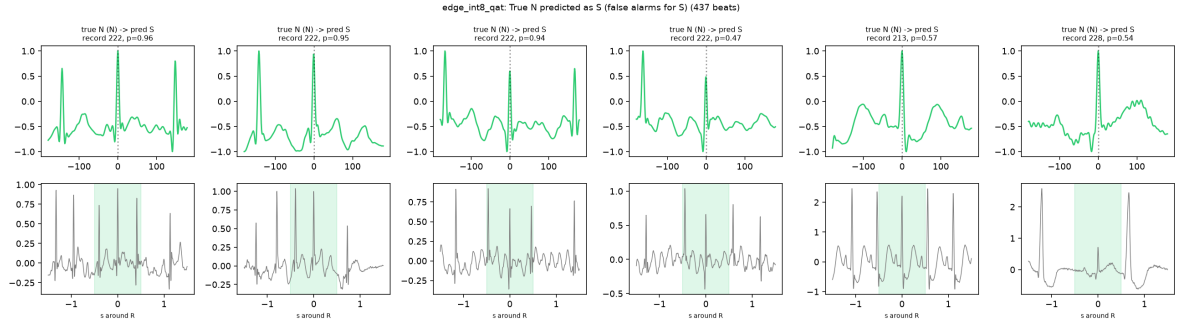
- 1738 beats were correct when clean and wrong with noise; 1277 went the other way.
- Where the noise pushes the predictions: V (57%), S (21%), N (17%), F (4%)

Class	Recall clean	Recall noisy	F1 clean	F1 noisy
N	0.949	0.944	0.955	0.948
S	0.044	0.040	0.066	0.059
V	0.767	0.696	0.587	0.548
F	0.000	0.000	0.000	0.000

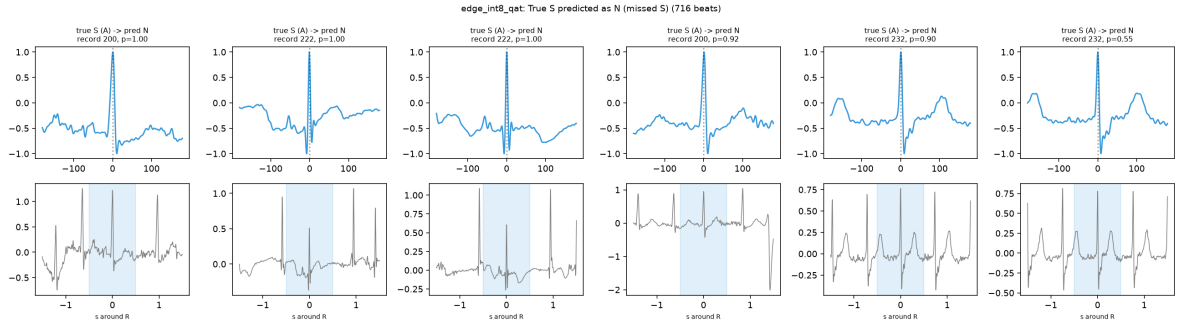
- `error_noise_examples.png`: beats that are correct when clean but wrong with noise



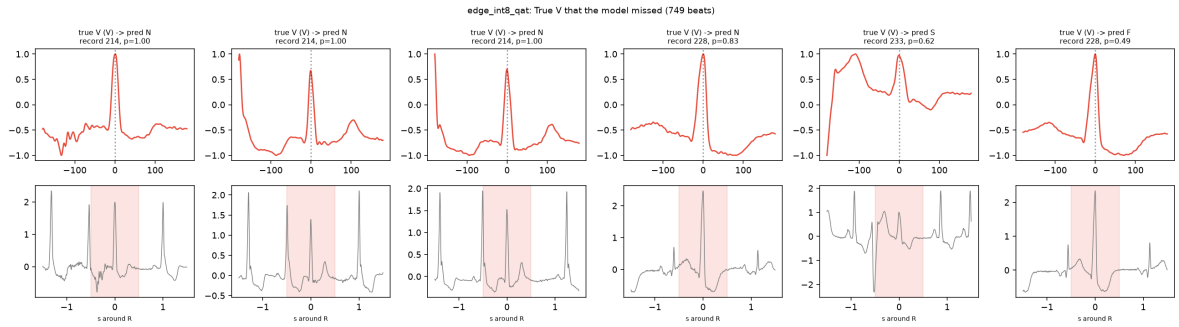
error_examples_F_missed.png



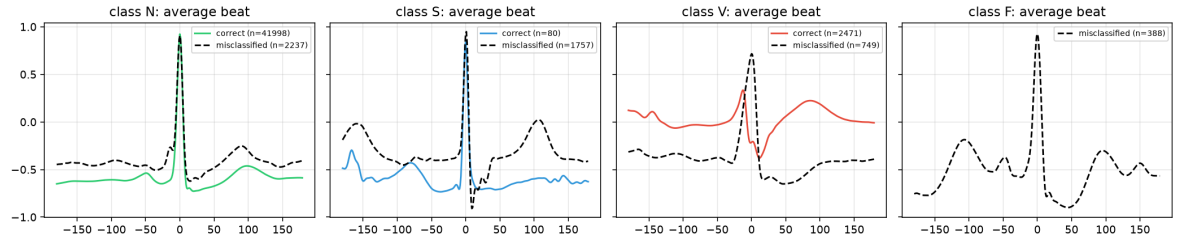
error_examples_N_to_S.png



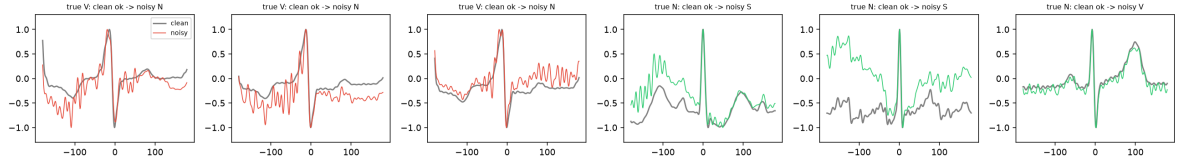
error_examples_S_to_N.png



error_examples_V_missed.png



error_mean_beats.png



error_noise_examples.png

The three records with the most errors (232, 213, 111) hold 55 % of all errors. Bundle-branch-block beats (L and R) are often predicted as V.

5.6 Success-criteria review

Criterion	Measured	Target	Verdict
Macro-F1 (INT8 emulation)	0.4019	>= 0.6	**Not achieved**
Macro-F1 above the best classical baseline	0.0007	>= 0.05	**Partially achieved**
Recall class V	0.7674	>= 0.8	**Partially achieved**
Recall class S	0.0435	>= 0.5	**Not achieved**
Recall class F	0.0000	>= 0.5	**Not achieved**
Macro-F1 drop from FP32 to INT8	0.0126	<= 0.03	**Achieved**
Macro-F1 drop at 0 dB noise	0.0220	<= 0.05	**Achieved**
Model size, INT8 (MB)	0.0253	<= 2.0	**Achieved**
MAX78002 synthesis passed	1.0000	>= 1	**Achieved**
Deployed ai8x model: macro-F1	0.4919	>= 0.6	**Partially achieved**
Deployed ai8x model: recall class V	0.8730	>= 0.8	**Achieved**
Deployed ai8x model: recall class S	0.1960	>= 0.5	**Not achieved**
Deployed ai8x model: recall class F	0.0077	>= 0.5	**Not achieved**
Process quality (no record in two splits; DS2 used once and logged)	see log		**Achieved**

Targets are those of section 1.4 (report_notes.yaml). “Partially achieved” means that the value lies between the target and the lower limit set for that criterion. Records 201 and 202 are the same subject (known limitation of the standard split).

6. Deployment: MAX78002 software synthesis

Synthesis result: PASSED

Generated C project

File	Present
cnn.c	yes
cnn.h	yes
main.c	yes
weights.h	yes
sampledata.h	yes
sampleoutput.h	yes

Model

- Input: [1, 360] (channels x samples), output: 4 classes
- Parameters stored on the chip (8-bit weights + biases): 25,268
- Estimated peak activation memory (8-bit, input + output of one layer): 11,520 bytes (estimate)

Resources reported by ai8xize

- Weight memory: 25,040 bytes of 2,396,160 bytes (1.0%)
- Bias memory: 228 bytes of 8,192 bytes (2.8%)
- Operations per inference: 1,420,688 (1,384,448 multiply-accumulate)

Layer	Operations	MACC
0	34,560	28,800
1	472,320	460,800
2	469,440	460,800
3	212,400	207,360
4	206,272	202,752
5	24,288	22,528

Layer	Operations	MACC
6	1,408	1,408

Quantization (8-bit) tensors from quantize.py

Tensor	Largest absolute value before quantization	Bits
conv1.op.weight	9.1198	8
conv1.op.bias	0.9842	8
conv2.op.weight	0.6224	8
conv2.op.bias	0.1568	8
conv3.op.weight	0.4254	8
conv3.op.bias	0.3689	8
conv4.op.weight	0.5275	8
conv4.op.bias	0.4306	8
conv5.op.weight	0.3989	8
conv5.op.bias	0.2146	8
conv6.op.weight	0.8108	8
conv6.op.bias	0.8479	8
fc.op.weight	0.4863	8
fc.op.bias	0.0151	8

Not measured

No MAX78002 board is used in this project, so real latency, energy and RAM use are not measured. The operation count above is a simulation-level quantity only.

Official 8-bit model from ai8x-training compared with the PyTorch INT8 emulation

Model	Evaluation set	Acc.	Macro-F1	F1 N	F1 S	F1 V	F1 F	Recall S	Recall V	Recall F
PyTorch INT8 emulation (main model)	validation records (4 of DS1)	0.8780	0.4551	0.934	0.098	0.578	0.211	0.235	0.575	0.133
ai8x 8-bit (deployed model)	validation records (4 of DS1)	0.9316	0.4933	0.966	0.312	0.670	0.025	0.376	0.652	0.067
PyTorch INT8 emulation (main model)	final test set DS2 (22 records, used once)	0.8967	0.4019	0.955	0.066	0.587	0.000	0.044	0.767	0.000
ai8x 8-bit (deployed model)	final test set DS2 (22 records, used once)	0.8758	0.4919	0.931	0.179	0.856	0.003	0.196	0.873	0.008

ai8x numbers are computed from the confusion matrices in max78002/quantization_output/ai8x_evaluation_*.log. The ai8x model was trained from scratch by ai8x-training, so it is a different training run, not a more exact copy of the emulated model.

Comparison of the two INT8 models (DS2). The deployed ai8x model reaches macro-F1 0.4919 and accuracy 0.8758; the PyTorch INT8 emulation reaches 0.4019 and 0.8967. The ai8x model is better on V (F1 0.856 versus 0.587) and S (0.179 versus 0.066) and slightly worse on N and accuracy: it finds more minority beats and pays with more false alarms (for example 1,635 N beats labelled F and 1,815 labelled S). F is still almost never found (recall 0.008).

It is not only a DS2 effect: on the validation records the ai8x model also scored higher (macro-F1 0.4933 versus 0.4551 for the emulation). The ai8x run is a separate training run (from scratch, checkpoint chosen on validation accuracy), not a more exact copy of the emulated model, and each model comes from a single run, so part of the difference may be run-to-run variation. We do not conclude that the toolchain or the chip is better.

The ai8x model scores about the same as the 2,900-parameter tiny CNN (0.4919 versus 0.4970), which suggests that the training recipe and the model selection matter more than the network size. The conclusion about S and F stays: they need timing information (RR interval) that a single 1-second window does not contain.

The emulation was kept as the main model because it was planned before DS2 was opened; the ai8x result was not used to change any decision.

7. Limitations and future work

- No board measurements: latency, energy and RAM use are not measured.
- No RR-interval feature: S and F beats are distinguished mainly by timing, which one 1-second window does not contain.
- Each INT8 model comes from one training run (one seed); the difference between the emulated and the ai8x model may partly be run-to-run variation.
- Very few F beats in the training records (15 in validation).

References

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Appendix A: Changes compared with the first submission

Submission 2 allows conclusions from Submission 1 to be revised when the experiments reveal problems. The changes are small; they are listed here with the reason.

#	Section	What changed	Why
1	1.2	Input described exactly (360 samples of lead MLII, filtered, normalised)	The assignment asks for the exact input of the model
2	1.4	Success criteria rewritten with numbers and one criterion per area	The first submission had only qualitative criteria; the assignment requires measurable ones and section 5.6 judges them
3	1.5, 2.1, 3.1	Records and subjects stated correctly: 48 records, 44 used (paced records 102, 104, 107, 217 excluded)	The report listed 44 records without saying why 4 of the 48 were left out
4	2.2, 3.6, 3.8	Beat counts explained (100,718 labelled, 100,667 in the final dataset); J corrected to j in class N	Two tables gave different totals; 51 beats were lost because their window leaves the recording; J belongs to class S
5	2.3 to 2.5	More figures from the EDA notebook; leakage experiment with numbers and a figure; record 116 and noise facts added	Evidence for statements already made
6	3.4	Pipeline order corrected; note about the zero-phase filter	Noise is added before filtering; a real device cannot use a zero-phase filter
7	3.6	Final class weights are softened (square root); balanced sampler tested and rejected with a result	Decided in the model selection of section 4.4, on validation data only
8	3.7	Full list of augmentations (gain, shift, baseline wander, noise)	Only two of the four were listed
9	3.9	Software of the Task 2 environment added	The versions must match what was actually run
10	References, typos	Two PhysioNet references added; typos fixed (1DD, Supraventricular)	The noise database and PhysioNet were used but not cited

Appendix B: every use of the final test set (DS2)

```
2026-11-04T12:59:11 | evaluate.py | clean evaluation of baseline, tiny_cnn_fp32, edge_fp32, edge_int8_ptq, edge_int8_qat
2026-11-04T13:04:05 | noise_test.py | noise robustness (bw, em, ma at [12, 6, 0] dB)
2026-11-04T13:04:23 | error_analysis.py | error analysis of edge_int8_qat
2026-11-04T15:03:45 | ai8x evaluate_ecg.sh | official 8-bit ai8x model on DS2 (once)
```

Seed 42; configuration file `best_config.yaml`. Patient-independent split by record (DS1 training/validation, DS2 final test).